

CONSTRUCTION OS

PRD - HSE Dashboard

Health, Safety & Environment Manager Role Workspace

Product purpose

Define the exact HSE Manager role shell, fixed safety dashboard, site-safety command surfaces, risk and permit control, field inspection, incident response, stop-work authority, workforce readiness, emergency readiness, environment and contractor HSE behavior. The experience is prevention-first and exception-first: know whether work is safe to start or continue, what is blocking it, and what requires immediate action.

Field	Value
Document	PRD_HSE_Dashboard
Version	1.0
Status	Implementation-ready role dashboard specification
Primary role	HSE Manager / Project HSE Manager
Secondary role compatibility	HSE Director, HSE Officer, Permit Issuer, Incident Investigator, Emergency Coordinator and HSE Auditor through permission-filtered views
Base requirements	PRD_HSE_Module v1.0 + PRD_HSE v1.0 + PRD_2 Role Centers + Complete Role & Permission Matrix + Approvals & Workflow Engine
Connected domains	Projects, Work Progress, HR, Assets, Inventory, Tasks, QA/QC, Documents, Contracts, Procurement, Calendar, Government & Legal, Finance, Notifications, Workflow
Scope	HSE role dashboard + exact HSE sidebar + site/risk/permit/incident/readiness workspaces
Date	2026-08-07

Permanent dashboard scope lock

The HSE Dashboard and HSE role sidebar shall contain only the elements explicitly defined in this PRD. Developers must not add, infer, duplicate, substitute or auto-expose extra dashboard widgets, modules, charts or shortcuts without an approved PRD revision. Platform Configuration may hide approved elements; it may not create unapproved ones.

1. Document Authority, Precedence and Product Position

Priority	Authority	Controls
1	Complete Role & Permission Matrix	HSE authority, project/site/record scope, protected medical/witness/investigation fields, deny-by-default behavior, separation of duties, revocation and audit.
2	PRD_HSE_Module v1.0	Operational HSE source of truth, dashboard KPI framework, site workspace, risk, permits, inspections, incidents, stop work, people readiness, emergency, environment, APIs and release phasing.
3	PRD_HSE v1.0	RAMS, LOTO, detailed HSE lifecycles, mobile/offline, security, analytics and HSE domain boundaries.
4	PRD_2 - Role Centers and Cues	Fixed permission-aware dashboard regions, primary cues, Work Inbox, operational content and recent activity.
5	Approvals & Workflow Engine v1.0	Decision routing, approval matrices, participants, work items, delegation, SLA, SoD, emergency override and source finalization.
6	Role Context and Shared Module Navigation	HSE shell remains active when authorized Project, Task, Document, HR, Asset, Calendar, QA/QC or Workflow content opens.
7	Platform UI/UX Architecture	Shared shell, Role Center, Work Inbox, Quick Create, progressive disclosure, responsive behavior, accessibility and motion.
8	This PRD	Exact HSE Manager dashboard composition, sidebar, site workspace and role-specific UX decisions.

1.1 Approval dependency resolution

Resolved dependency

The HSE source PRDs describe organization-wide approval orchestration as deferred or integration-ready. The platform Approvals & Workflow Engine now exists and governs routing, participants, decisions, delegation, SLA, escalation, overrides and source-finalization requests. HSE still owns hazards, risk assessments, permits, stop-work decisions, incidents, investigations, emergency/environmental records and final HSE business transitions.

1.2 Executive purpose

- Give HSE leadership one calm command surface for immediate safety risk, today's high-risk work, permits, incidents, inspections, actions, stop-work and readiness.
- Prioritize prevention and active blockers before retrospective analytics.
- Keep company, project, site, contractor and date context explicit.
- Make hazard, permit, inspection, incident and stop-work actions reachable within two navigation decisions.
- Prevent dashboard convenience from bypassing risk hierarchy, safety gates, permit validation, confidentiality, evidence, workflow, audit or source ownership.

Final product position

HSE Manager opens Dashboard -> sees critical safety state -> confirms whether high-risk work may proceed -> resolves permits, risks, incidents, failed inspections and holds -> verifies corrective actions -> opens the authoritative HSE record -> completes the controlled action. The dashboard is a permission-safe operational projection, not a second HSE database.

2. Goals, Non-Goals and Role Scope

In scope	Out of scope
Exact HSE role shell, sidebar and fixed dashboard	A separate safety application that copies HR, Assets, Tasks, Projects, QA/QC or Documents records
Company/project/site/contractor/date scoped HSE control	Automatic visibility of all companies, sites, incidents or confidential investigations
Critical alerts, today view, prevention/readiness and operational exceptions	Arbitrary dashboard drag-and-drop or user-added KPI widgets
Hazards, risk assessments, RAMS, permits, inspections, observations, incidents, actions and stop-work navigation	Replacing Work Progress as physical work/production truth or QA/QC as quality acceptance truth
Worker eligibility, induction, competence and PPE readiness projections	Replacing HR employment, training master or detailed medical records
Asset safety readiness projections	Replacing Assets maintenance/certification/equipment lifecycle
Environment, emergency and contractor HSE	Jurisdiction-specific legal advice or hard-coded statutory filing logic
Workflow decision projections where configured	A parallel HSE-specific approval engine

2.1 Role scope model

```
AUTHENTICATED USER
-> Active Company
    -> Active Role: HSE Manager
        -> Authorized Project / Site Scope
            -> Optional Contractor / Date Filters
                -> HSE Dashboard / Site HSE / Shared Source Pages
```

- Company/project/site/location/contractor/record scope is enforced server-side.
- Filters narrow authorized data; they never expand access.
- Medical detail, witness statements, investigation analysis, severe-event governance and stop-work release remain independently permission controlled.
- Multi-role users remain in active HSE role until they explicitly switch roles.

3. Application Shell, Header and Exact Sidebar

3.1 Persistent header

```
HSE
[ Company v ] [ Project v ] [ Site v ] [ Contractor v ] [ Date v ]
[ Global Search ] [ + New ] [ Notifications ] [ Profile / Role ]
```

- Company/project/site changes re-resolve authorization, dashboard aggregates, saved views and sidebar visibility.
- Contractor and date are narrowing filters only.
- Global Search is permission-safe and opens source records inside the active HSE shell.
- Notifications deep-link to the source record or Workflow work item without changing active role.

3.2 Exact approved sidebar

```
HSE
  Dashboard
  Projects & Sites

SAFETY PLANNING
  Safety Plans
  Hazards & Risks
  Risk Assessments
  Method Statements / RAMS

WORK CONTROL
  Permits to Work
  Inspections
  Safety Observations
  Stop-Work Orders

INCIDENTS
  Incidents & Near Misses
  Investigations
  Corrective Actions

PEOPLE & READINESS
  Inductions
  Toolbox Talks
  Training & Competence
  PPE

EMERGENCY
  Emergency Management

ENVIRONMENT
  Environmental Management
  Chemicals / Hazardous Substances

CONTRACTORS
  Contractor HSE

  Occupational Exposure

REPORTING
  Reports

SETTINGS
  HSE Settings
```

- Group labels are visual separators, not routes.
- Unauthorized or configuration-disabled items disappear with clean reflow.
- Projects, Tasks, Documents, HR, Assets, Inventory, QA/QC, Contracts, Calendar, Government & Legal, Finance and Workflow remain shared source surfaces rather than duplicate HSE sidebar modules.
- Opening an authorized shared record preserves HSE shell and role context.

4. Full HSE Dashboard Architecture Map

```
HSE DASHBOARD
|
|-- Header / Scope Controls
|   |-- Company / Project / Site
|   |-- Contractor / Date
|   |-- Search / + New / Notifications
|
|-- Primary Cues
|   |-- Critical Incidents
|   |-- Active Permits
|   |-- Critical Risks
|   |-- Open Stop-work Orders
|   |-- Inspections Due
|   |-- Overdue HSE Actions
|   |-- Workers Not Eligible
|   |-- Emergency Readiness Gaps
|
|-- Immediate Attention
|-- Today
|-- Risk & Safety Readiness
|-- Permits & High-Risk Work
|-- Incidents & Near Misses
|-- Inspections & Observations
|-- Corrective Actions
|-- Stop-work / Safety Holds
|-- Workforce Readiness
|-- Emergency Readiness
|-- Environmental Status
|-- Contractor HSE
|-- Work Inbox
|-- Recent Activity
```

Layout lock

These are the complete approved HSE Dashboard regions. There is no generic Finance, Sales, HR, productivity-ranking or arbitrary analytics area. Reports may provide deeper HSE analytics, but new dashboard widgets require an approved PRD revision.

5. Primary Cues - Exact Set

Cue	Definition	Click-through
Critical Incidents	Authorized open incidents at configured critical/severe threshold or emergency response state.	Incidents filtered to critical/open.
Active Permits	Authorized permits currently Active and within valid server-authoritative time window.	Permits filtered to Active.
Critical Risks	Authorized hazard/risk records with current critical risk or configured unacceptable residual risk.	Risk register filtered to critical.
Open Stop-Work Orders	Authorized stop-work/safety-hold records not governed-released or closed.	Stop Work filtered to active/open.
Inspections Due	Authorized safety inspections due today or overdue and not final/closed.	Inspections filtered to due/overdue.
Overdue HSE Actions	Authorized corrective/preventive HSE actions beyond governed due date and not verified/closed.	Corrective Actions filtered to overdue.
Workers Not Eligible	Authorized workers in selected scope failing configured induction/competence/fitness/PPE eligibility gates.	Worker readiness filtered to not eligible.
Emergency Readiness Gaps	Authorized unresolved emergency plan/contact/equipment/drill/readiness deficiencies.	Emergency Management filtered to gaps.

- Each cue shows current count/value, semantic status, last-calculated/freshness data where computed and drill-through.
- Counts reconcile with the exact filtered authoritative register and permission scope.
- Critical state is never conveyed only by color.
- Cue calculations must not classify individual employees as good or poor performers.

6. Immediate Attention

IMMEDIATE ATTENTION
+-- Critical Incidents
+-- Active Emergency
+-- Critical Stop-Work Orders
+-- Expired High-Risk Permits
+-- Unsafe Assets
+-- Critical Uncontrolled Risks
+-- Permit Conflicts
`-- Critical Overdue Actions

6.1 Behavior

- Immediate Attention is the highest-priority dashboard region and remains above passive trend content.
- Critical items remain visible until source resolution or an explicit acknowledgement where acknowledgement is allowed.
- Acknowledgement never changes the underlying HSE business state.
- Each item shows source record, company/project/site/location, severity/status, owner/responsible party, due/expiry time and required action.
- Active emergencies and stop-work orders cannot be hidden by ordinary user preferences.

Critical safety rule

A mandatory blocker or active safety hold overrides any favorable readiness percentage, schedule pressure or commercial status.

7. Today - Operational Safety Control

```
TODAY
+-- Active Permits
+-- Planned High-Risk Work
+-- Inspections Due
+-- Toolbox Talks
+-- Pending Inductions
+-- Permit Expiries
+-- Planned Emergency Activities
`-- Safety Actions Due
```

- Today is site-time-zone aware and follows the active company/project/site scope.
- High-risk work must show the linked risk/RAMS/permit readiness necessary to start.
- Permit expiry uses server-authoritative validity, not browser time alone.
- Planned activities deep-link to HSE or Calendar source records while HSE retains control of HSE status transitions.
- Today never infers that work is safe merely because it is scheduled.

7.1 Today list row

Field	Requirement
Time / window	Local site time plus expiry/due state.
Activity	High-risk work, permit, inspection, induction, toolbox or action.
Scope	Project / site / exact location / work package where available.
Responsible	Authorized issuer, receiver, inspector, supervisor or action owner.
Readiness	Explicit Ready / Blocked / Warning / Pending state with blocker explanation.
Action	Open source record or perform permitted controlled action.

8. Risk & Safety Readiness

```
RISK & SAFETY READINESS
+-- Critical Risks
+-- High Residual Risks
+-- Risk Reviews Overdue
+-- Controls Not Verified
+-- Safety Plan Status
+-- Worker Eligibility
+-- Asset Safety Readiness
`-- Emergency Readiness
```

8.1 Readiness rule

- Readiness is calculated from explicit rules and latest authoritative data.
- Computed readiness shows data freshness / last-calculated timestamp.
- Every failed readiness state identifies exact unmet gates.
- Historical readiness snapshots remain reproducible after source data changes.
- No single percentage may conceal a critical blocker.

8.2 Work-start gate concept

```
WORK / ACTIVITY
-> HSE Plan / applicable rules
-> Hazard / Risk Assessment
-> RAMS / method readiness
-> Required Permit(s)
-> Worker eligibility
-> Asset safety readiness
-> Isolation / weather / emergency checks
-> NO ACTIVE STOP-WORK
=> READY TO START or BLOCKED WITH EXACT REASONS
```

9. Hazards, Risk Assessments and Control Hierarchy

```
Hazard
-> Initial Risk
  -> Controls
    -> Residual Risk
      -> Review / Release
        -> Active Control
          -> Monitoring / Revision
```

9.1 Required control hierarchy

1. Elimination
2. Substitution
3. Engineering Controls
4. Administrative Controls
5. PPE

- Risk templates must explicitly capture the control hierarchy and must not default to PPE when stronger controls are available.
- High-consequence hazards relying only on administrative controls or PPE must be visibly identified according to HSE policy.
- Issued/released risk assessment revisions are immutable; changes create controlled revisions.
- Risk revision during an active permit preserves the permit's original referenced revision and triggers configured review/revalidation behavior.

9.2 Risk Assessment workspace

```
RISK ASSESSMENT
+-- Overview / Scope / Activity
+-- Location
+-- Hazards
+-- Initial Risk
+-- Controls / Hierarchy
+-- Responsible Parties
+-- Residual Risk
+-- Workers / Competence
+-- Assets / Equipment
+-- PPE Requirements
+-- Emergency Controls
+-- Documents
+-- Linked RAMS / Permits
+-- Reviews / Revisions
`-- Activity
```

10. Method Statements / RAMS

```
RAMS
+-- Risk Assessment
+-- Method Statement
+-- Work Sequence
+-- Responsibilities
+-- Equipment / Materials
+-- Personnel / Competence
+-- Permits
+-- PPE
+-- Safety / Environmental Controls
+-- Emergency Controls
+-- Quality / Inspection Points
+-- Controlled Drawings / Specifications
+-- Worker Acknowledgements
`-- Revision History
```

10.1 Canonical readiness

Incomplete -> Ready for Review -> Issued -> Active -> Expired / Superseded / Suspended

- RAMS references exact controlled document revisions in Documents.
- Expired, superseded or suspended RAMS must block configured work-start and permit gates immediately.
- Worker acknowledgement records identity, organization where applicable, language/version and timestamp.
- HSE may display Documents records and acknowledgement state but does not become the document source of truth.

11. Permits & High-Risk Work

PERMITS & HIGH-RISK WORK
+-- Active
+-- Ready for Issue
+-- Expiring
+-- Expired
+-- Suspended
+-- Conflicts / SIMOPS
+-- Revalidation Required
+-- Closure Due
`-- Closed

11.1 Supported permit classes

Examples	Examples
Hot work	Work at height
Confined space	Excavation
Electrical isolation / LOTO	Lifting / crane operation
Demolition	Temporary works
Road closure / live services	Pressure testing / radiography
Chemical work	Fire-system isolation
Environmental discharge	Night work / custom

11.2 Lifecycle

Draft -> Requested / Prepared -> Ready for Issue -> Issued -> Active -> Suspended -> Revalidated -> Closed
-> Archived
Alternative: Rejected / Expired / Cancelled / Revoked / Emergency Closed

- Expired permits automatically leave Active state and publish a blocking event.
- Shift extension or responsibility transfer requires revalidation and explicit acceptance where configured.
- Conflicting simultaneous operations are detected by permit type, location and time rules.
- Backdated permit activation is prohibited.

12. Permit Page and Activation Gates

```

PERMIT TO WORK
+-- Overview / Type / Number
+-- Project / Exact Location
+-- Work Description / Contractor
+-- Issuer / Receiver / Supervisor
+-- Workers
+-- Assets / Equipment
+-- Risk Assessment
+-- RAMS / Method Statement
+-- Isolation / LOTO
+-- Tests / Readings / Weather
+-- PPE
+-- Emergency Arrangements
+-- SIMOPS / Conflicts
+-- Evidence / Documents
+-- Signatures
+-- Revalidation / Suspension History
+-- Closure / Handback
`-- Activity

```

12.1 Activation gates

```

VALID RISK ASSESSMENT
+ VALID RAMS / METHOD
+ COMPETENT & INDUCTED WORKERS
+ CURRENT TRAINING / CERTIFICATES
+ VALID ASSET INSPECTIONS
+ CONFIRMED ISOLATION
+ ACCEPTABLE WEATHER / AREA
+ EMERGENCY ARRANGEMENTS READY
+ NO CONFLICTING PERMIT
+ REQUIRED DOCUMENTS PRESENT
+ NO STOP-WORK HOLD
=> SERVER AUTHORITATIVE ACTIVATION

```

- Permit activation is always server-authoritative; no client-only or offline activation.
- Closure confirms area safe, housekeeping/restoration complete, isolations restored/transferred and outstanding hazards recorded.
- A stop-work order suspends affected permits immediately.
- Permit activation/closure may use Workflow where configured, but HSE owns permit validation and final permit state.

13. Safety Inspections

```
SAFETY INSPECTIONS
+-- Due Today
+-- Upcoming
+-- In Progress
+-- Failed
+-- Critical Findings
+-- Corrective Action Required
+-- Verification Required
+-- Overdue
`-- Closed
```

13.1 Inspection flow

```
Inspection
-> Versioned Checklist
-> Evidence / Measurements
-> Findings
  +-- Pass -> Close / accepted result
  `-- Fail -> Hazard / Observation / Corrective Action / Stop Work
        -> Correction
        -> Verification
        -> Close
```

- Inspection templates are versioned; final inspection history preserves the exact template revision.
- Critical failures trigger configured consequences and may issue alerts, actions, hazards or stop-work.
- Closed inspections remain historically reproducible and are not edited in place.
- Inspection evidence keeps actor/time/source integrity; offline capture remains pending until server acknowledgement.

14. Safety Observations and Rapid Field Reporting

```
SAFETY OBSERVATION
+-- Unsafe Condition
+-- Unsafe Act
+-- Positive Observation
+-- Improvement Opportunity
`-- Immediate Danger
```

14.1 Observation flow

Capture -> Triage -> No Action / Hazard / Corrective Action / Stop Work -> Verification -> Closed

- Mobile observation capture minimizes form friction while preserving scope, location, reporter, time and evidence.
- Immediate danger can escalate to stop-work without waiting for a full investigation form.
- Positive observations may be reported as leading indicators but must not automatically score individual employees.
- Observation conversion to hazard/action/stop-work uses stable source links; records are not duplicated.

15. Incidents & Near Misses

```
INCIDENT CONTROL
+-- Critical Incidents
+-- Recordable Incidents
+-- Lost-Time Incidents
+-- Medical Treatment
+-- First Aid
+-- Near Misses
+-- Dangerous Occurrences
+-- Property Damage
+-- Environmental Events
+-- Under Investigation
`-- Closure Pending
```

15.1 Lifecycle

```
Initial Report -> Triage -> Immediate Response -> Investigation -> Root Cause -> Corrective / Preventive
Actions -> Effectiveness Verification -> Closure
```

- Initial reporting supports fast minimum-safe capture; deeper investigation may follow.
- Severity/response triage is controlled and auditable.
- Critical incidents trigger mandatory notification/escalation events according to policy.
- Final investigations and closed incidents remain immutable; correction uses reopening/addendum/superseding record.

16. Incident Page and Investigation

INCIDENT
+-- Overview / Classification
+-- Project / Site / Location / Activity
+-- People Involved
+-- Witnesses
+-- Assets / Equipment
+-- Immediate Response
+-- Evidence
+-- Investigation
+-- Root Cause
+-- Corrective / Preventive Actions
+-- Stop-Work Links
+-- Documents
+-- Notifications / Escalation
+-- Closure
`-- Activity

16.1 Confidentiality zones

Zone	Examples	Default behavior
Operational HSE	Incident class, location, basic response, safety controls.	Visible only within authorized HSE/project scope.
Investigation restricted	Root-cause analysis, witness statements, sensitive evidence.	Explicit record/field capability required.
Medical restricted	Detailed medical data / diagnosis.	Remains HR / Occupational Health; HSE receives minimum operational fitness/clearance projection.
Legal restricted	Privileged legal analysis / regulatory strategy.	Remains Government & Legal / Legal domain.

- A user who can view the incident shell must not automatically receive witness, medical or legal detail.
- Evidence originals retain integrity metadata and source links.
- Final findings preserve actor, record version, effective time and correlation ID.

17. Corrective and Preventive Actions

HSE ACTIONS
+-- New
+-- Assigned
+-- In Progress
+-- Overdue
+-- Ready for Verification
+-- Effective
+-- Ineffective / Reopened
`-- Closed

17.1 Ownership separation

Capability	Owner	Behavior
Required HSE outcome	HSE	Defines required result, due state, evidence and verification criteria.
Execution task	Tasks	Owens assignee, execution workflow, comments and task status.
Verification	HSE	Checks completion/evidence against HSE requirement.
Effectiveness review	HSE	Confirms whether the control corrected/prevented recurrence.
Closure	HSE	Finalizes only after source validation; may reopen if ineffective.

No automatic closure

Task complete is not equal to HSE action effective. HSE verification/effectiveness remains separate and may reopen or create further action.

18. Stop-Work Orders and Safety Holds

```
STOP WORK
+-- Active Holds
+-- Critical Holds
+-- Under Review
+-- Release Requested
+-- Released
`-- History
```

18.1 Operational flow

```
Unsafe condition / event
-> STOP WORK ISSUED
-> Affected work blocked
-> Affected permits suspended
-> Corrective controls
-> Verification
-> Governed release
-> Work may resume
```

- Stop-work authority must be fast, visible, scoped and auditable.
- A stop-work order identifies affected project/site/location/work package/asset/permit scope.
- Release authority is separate where policy requires; the issuer cannot automatically self-release.
- Workflow may route release/exception decisions, but HSE owns the hold and final release validation.
- Offline users may issue a stop-work notice where configured, but final release cannot occur offline.

No silent bypass

There is no generic administrator Skip Stop Work or Resume Work control. Every release or emergency override requires explicit capability, reason, evidence and audit.

19. Workforce Readiness

```
WORKFORCE READINESS
+-- Induction Status
+-- Training / Competence Validity
+-- Permit Eligibility
+-- Toolbox Participation
+-- PPE Compliance
`-- Operational Fitness Status
```

19.1 Source boundary

```
Worker / User
+-- Identity / Employment ----- HR / User Profile
+-- Training / Certification Master - HR
+-- Medical Detail ----- HR / Occupational Health
+-- Operational Fitness Projection - HR -> HSE
+-- Inductions ----- HSE
+-- Toolbox Talks ----- HSE
+-- PPE Requirements ----- HSE
`-- Worker Eligibility ----- HSE calculated projection
```

- HSE may consume active employment, competence/certification and operational fitness status but not detailed medical records.
- Eligibility is calculated by explicit configured rules for activity/site/permit context.
- Eligibility failure states must state the exact missing requirement without exposing unrelated private data.
- Worker eligibility is an operational safety gate, not an employee performance score.

20. Inductions, Toolbox Talks, Competence and PPE

20.1 Inductions

- Induction event records content/revision, site/project, participant, attendance, acknowledgement and expiry/validity where configured.
- Expired/missing induction may block configured worker or permit eligibility.

20.2 Toolbox Talks

- Toolbox talk records topic, scope, work package/site, responsible supervisor, participant attendance, acknowledgement and follow-up.
- Participation contributes to leading indicators without becoming an automatic worker rating.

20.3 Training & competence

- HR remains the master for training/certification records; HSE defines activity/site competence requirements and evaluates current status.
- Permit-type and site-specific competence can restrict issuer/receiver/worker eligibility.

20.4 PPE

- HSE defines PPE requirements and compliance context; Inventory remains authoritative for PPE stock/issue quantity where used.
- PPE is the last level of the control hierarchy and must never replace stronger available controls in risk design.

21. Emergency Readiness and Emergency Mode

```
EMERGENCY MANAGEMENT
+-- Current Emergency Status
+-- Emergency Plans
+-- Emergency Contacts
+-- Muster Points
+-- Response Teams
+-- Emergency Equipment
+-- Drills
+-- Readiness Gaps
`-- Event History
```

21.1 Emergency lifecycle

```
Normal -> Emergency Activated -> Response -> Muster / Control -> Stabilized -> Resolution -> Post-Event
Review
```

- Active emergency state is always visible on the HSE Dashboard for affected scope.
- Emergency contacts/plans must remain quickly accessible; cached emergency information may be available offline according to Mobile/Offline policy.
- Emergency resolution is server-authoritative and cannot be finalized offline.
- Emergency override follows configured scope, duration, evidence, oversight and post-event review; it never erases skipped controls or prior decisions.

22. Environmental Management and Hazardous Substances

```
ENVIRONMENT
+-- Environmental Aspects / Risks
+-- Monitoring
+-- Environmental Events / Spills
+-- Waste
+-- Exceedances
+-- Water / Energy / Fuel Trends
+-- Chemicals / Hazardous Substances
`-- Corrective Actions
```

22.1 Chemical register

- Chemical/hazardous-substance records contain operational HSE context and link to controlled SDS/document revisions.
- Storage quantity/location remains Inventory authority where applicable; HSE owns hazard/control context.
- Environmental exceedances and spills can trigger alerts, corrective action, stop-work/hold or Government & Legal linkage according to rules.
- External environmental sensors are future/optional integrations and are not required for core HSE operation.

23. Contractor HSE

```
CONTRACTOR HSE
+-- HSE Profile
+-- Qualification
+-- Worker Eligibility
+-- RAMS
+-- Permits
+-- Inspections
+-- Incidents / Violations
+-- Corrective Actions
+-- Induction Compliance
`-- HSE Performance
```

- Contractor HSE views use the authoritative contractor/supplier identity and do not create a duplicate company master.
- Performance is derived from permission-safe HSE evidence such as incidents, permit violations, overdue actions and induction compliance.
- Contractor HSE access is restricted to authorized company/project/site scope.
- External contractor portal participation, when enabled later, exposes only explicitly shared HSE actions/records and never internal investigation detail.

23.1 Contractor scorecard guardrail

No hidden worker scoring

Contractor aggregate HSE performance may support sourcing and governance decisions, but the dashboard must not automatically classify individual employees as good or poor performers.

24. Site HSE Workspace

```
SITE HSE
+-- Overview
+-- Safety Readiness
+-- HSE Plan
+-- Hazards
+-- Risk Assessments
+-- RAMS
+-- Permits
+-- Inspections
+-- Observations
+-- Incidents
+-- Corrective Actions
+-- Stop Work
+-- Workers / Eligibility
+-- Inductions
+-- Toolbox Talks
+-- Assets / Equipment Readiness
+-- Emergency
+-- Environment
+-- Contractors
+-- Documents
+-- Activity
```

- Every active project site has one HSE operational workspace combining current readiness, active controls, workforce eligibility, emergency status and unresolved risk.
- All sections are projections or direct source records; the site workspace is not a duplicate data domain.
- Project/site scope persists across deep links and back navigation.
- Critical blockers and emergency/stop-work state remain visually persistent across the workspace.

25. Work Inbox, Recent Activity and Notifications

25.1 Work Inbox

Inbox type	HSE examples	Primary action
Task	Corrective task, emergency preparation task.	Open Tasks source.
Decision	Permit/safety review, stop-work release, investigation/closure decision where configured.	Open Workflow decision.
Reading / acknowledgement	HSE instruction, plan revision, emergency notice, RAMS acknowledgement.	Open exact document/notice revision and acknowledge.
Clarification	Returned assessment/permit/incident submission.	Respond/resubmit.
Exception	Permit conflict, failed sync, asset safety non-compliance, blocked gate.	Resolve/escalate source record.
Deadline	Permit expiry, overdue action, training/induction expiry, drill due.	Open source record.

25.2 Recent Activity

- Permission-filtered timeline of HSE events relevant to the active company/project/site scope.
- Activity rows identify source module, record, action, actor, effective/server time and status without leaking restricted fields.
- Critical informational notifications remain separate from actionable Work Inbox items.

25.3 Notification-ready events

Critical Incident Reported
 Emergency Activated
 Stop Work Issued
 Permit Expiring / Suspended
 High-Risk Work Starting
 Inspection Failed
 Corrective Action Overdue
 Training / Induction Expiring
 Asset Safety Non-Compliant
 Environmental Limit Exceeded
 Emergency Drill Due

26. Search, Filters, Saved Views and Quick Create

26.1 Dashboard filters

Company
Project
Site
Contractor
Date

- Dashboard filter context is preserved in cue drill-down and back navigation.
- Filters are URL-serializable where platform patterns permit.
- Saved views are private by default and cannot broaden permissions.

26.2 Searchable HSE records

- Sites, plans, hazards, assessments, RAMS/method links, permits, inspections, observations, incidents, investigations, actions, holds, inductions, toolbox talks, emergencies, environmental events, chemicals and contractor HSE records.

26.3 Quick Create

+ New
+-- Hazard
+-- Risk Assessment
+-- Permit to Work
+-- Safety Inspection
+-- Safety Observation
+-- Incident / Near Miss
+-- Corrective Action
+-- Stop-Work Order
+-- Induction / Toolbox Talk (where authorized)
`-- Environmental Event

- Quick Create exposes only authorized and configuration-enabled record types relevant to active context.
- Creation pre-populates company/project/site scope where known but never fabricates unavailable source data.

27. Cross-Module Ownership and Integration

Domain	System of record	HSE dashboard behavior
Projects / Sites	Projects	Reads project/site/work-package context and safety projections; never changes project ownership.
Work Progress	Work Progress	Publishes readiness/hold gates; Work Progress consumes them but cannot edit HSE source records.
HR / User Profile	HR / Identity	Reads identity, employment, competence and operational fitness projection; detailed medical remains restricted.
Assets	Assets	Reads asset identity, inspection/certification/technical state; HSE may issue/request hold and incident links.
Inventory	Inventory	Reads PPE availability/issue and hazardous-material quantity/location where authorized; HSE owns safety requirements.
Tasks	Tasks	HSE defines corrective outcome/verification; Tasks owns execution.
QA/QC	QA/QC	Links safety-impacting quality issues without merging quality and safety records.
Documents	Documents	Stores files/revisions; HSE stores stable references and exact revision context.
Contracts	Contracts	Reads HSE obligations and deadlines; legal terms remain Contracts.
Procurement	Procurement	Reads contractor/supplier purchasing context; quality/safety outcomes return as linked evidence.
Calendar	Calendar	Displays HSE date projections/deep links; HSE owns HSE scheduling/status changes.
Government & Legal	Government & Legal	Links official permits/regulatory submissions/outcomes; HSE does not hard-code legal filing truth.
Finance	Finance	May display authorized cost-impact references; never posts accounting entries.
Notifications	Event Centre	HSE emits facts; Event Centre controls delivery channels/policy.
Workflow	Workflow Engine	Owens decision routing; HSE owns source record and final HSE state.

Permanent source-of-truth rule

HSE owns operational safety and environmental control records. Connected modules retain ownership of people, assets, stock, work completion, quality acceptance, files, legal terms and accounting. Cross-module links use stable references and permission-safe projections.

28. Approvals, Workflow and Emergency Override

```
HSE SOURCE RECORD
-> Submit / Request Decision
  -> Workflow Selection
    -> Participants / Work Items / SLA
      -> Decisions
        -> Source Finalization Request
          -> HSE Source Validation
            -> HSE State Transition
              -> Workflow Complete
```

28.1 Candidate HSE workflow decisions

- HSE Plan or risk-assessment issue/release where configured.
- Permit issue/activation or exceptional permit decision where company policy requires review.
- Stop-work release or hold override.
- Critical incident closure / final investigation approval where configured.
- Environmental exception or regulatory-sensitive action.
- Emergency override and mandatory independent post-event review.

28.2 Workflow rules

- Workflow owns routing, participants, delegation, decisions, SLA, escalation and decision audit.
- HSE owns record content, safety validation and final HSE state.
- Workflow approval is not treated as final HSE state until HSE confirms source finalization.
- HSE Critical confidentiality may use minimum decision projections so an approver receives only fields required to decide.
- No administrator inherits business approval or stop-work release authority merely through platform/workflow administration.

29. Role Context, Platform Configuration and Release Phasing

29.1 Role-context persistence

Opened content	Required shell behavior
Project / Site	Authorized project/site opens inside HSE shell; HSE sidebar/header remain active.
Task	Corrective task opens without switching to generic Task/Project role shell.
Document	Controlled HSE/RAMS/SDS document opens without switching role context.
Employee	Authorized eligibility/profile context opens without switching to HR dashboard.
Asset	Authorized asset safety context opens without switching to Assets dashboard.
QA/QC / Contract / Procurement / Calendar	Authorized linked source page opens permission-safe while HSE shell persists.
Workflow	Approval/decision opens inside current HSE role context where platform pattern permits.

Navigation rule

Authenticated active role/workspace resolves the shell. Route resolves page content. Opening a shared module must never silently switch the user into another role.

29.2 Platform Configuration

- Every approved sidebar item, dashboard region, field/action/report and HSE feature is configuration-aware.
- Disabled functionality disappears from navigation, search, APIs and layout with no dead space.
- Configuration cannot grant permission, bypass HSE lifecycle/gates/workflow, or introduce unapproved dashboard content.

29.3 Delivery phasing

Phase	Approved scope
Operational Foundation	Dashboard/Site HSE, hazards/risk, RAMS links, permits, inspections/observations, incidents/investigations/actions, stop work, inductions/toolbox, emergency contacts, Documents/Tasks/Assets links, responsive mobile, audit/core reports.
Advanced Control	PPE/worker eligibility, Contractor HSE, environment, chemical register, drills/advanced emergency, advanced analytics, offline operations, Work Progress safety gates, Calendar integration.
Future optional	IoT sensors, wearables/geofencing, automatic muster, telematics, AI-assisted hazard/PPE detection, predictive analytics, external regulatory/video integrations.

30. Data Model, Dashboard Read Models and Integrity

30.1 Representative HSE source entities

Family	Representative entities
Site / planning	HSESite, HSEPlan, Hazard, HazardCategory, RiskAssessment, RiskAssessmentHazard, RiskControl, MethodStatementSafetyLink.
Permits	PermitType, PermitToWork, PermitWorker, PermitAsset, PermitIsolation, PermitTest.
Field control	SafetyInspection, InspectionTemplate, InspectionResponse, SafetyObservation, StopWorkOrder.
Incidents / actions	Incident, IncidentPerson, IncidentWitness, IncidentAsset, IncidentEvidence, IncidentInvestigation, RootCause, CorrectiveAction, EffectivenessReview.
People	Induction, InductionAttendance, ToolboxTalk, ToolboxAttendance, CompetenceRequirement, WorkerEligibility, PPERequirement, PPEAssignment.
Emergency / environment	EmergencyPlan, EmergencyContact, EmergencyDrill, MusterPoint, EnvironmentalAspect, EnvironmentalMonitoring, EnvironmentalEvent, WasteRecord, ChemicalRegister.
Contractor / shared	ContractorHSEProfile, ContractorHSEEvaluation, OccupationalExposure, HSEDocumentLink, HSEActivity, HSEMetricSnapshot.

30.2 Core integrity rules

- Stable UUID/business identifier, tenant/company and project/site scope on governed records.
- Immutable issued/final records; no hard deletion of governed history.
- Optimistic version control for mutable aggregates and exact effective-dated configuration snapshots.
- Stable source correlation IDs and transactional outbox for events.
- Dashboard read models are read-only projections; no dashboard-owned duplicate lifecycle state.

31. Frontend Architecture, State and Logical APIs

31.1 Component architecture

```
HSEShell
+-- HSEDashboardPage
|   +-- PrimaryCues
|   +-- ImmediateAttention
|   +-- TodayPanel
|   +-- RiskReadinessPanel
|   +-- PermitHighRiskPanel
|   +-- IncidentPanel
|   +-- InspectionObservationPanel
|   +-- CorrectiveActionsPanel
|   +-- StopWorkPanel
|   +-- WorkforceReadinessPanel
|   +-- EmergencyReadinessPanel
|   +-- EnvironmentalStatusPanel
|   +-- ContractorHSEPanel
|   +-- WorkInbox
|   `-- RecentActivity
+-- SiteHSEPage
+-- RiskAssessmentWorkspace
+-- PermitWorkspace
+-- IncidentResponseWorkspace
+-- StopWorkWorkspace
`-- HSEQuickCreate
```

31.2 State rules

- Server state is authoritative; client state stores scope filters, UI preferences and drafts only.
- Query/cache keys include tenant, company, user/effective authorization, active role, project/site, contractor/date and authorization version.
- Official transitions wait for server authority; optimistic UI is limited to reversible drafts.
- Late responses from old company/site/role context cannot overwrite active dashboard state.
- Event invalidation updates exact affected aggregates; reconciliation repairs drift.

31.3 Logical API boundaries

Capability	Purpose
GET HSE Dashboard	Scoped cues, immediate attention, today, readiness, operational panels, Work Inbox and activity.
GET Sites / Site HSE	Site register/readiness summaries and full Site HSE projection.
GET/POST Hazards / Risk Assessments	Source HSE risk operations and immutable release/revision commands.
GET/POST Permits	List, create, issue, activate, suspend, revalidate and close through server-authoritative HSE APIs.
GET/POST Inspections / Observations	Field-control records and result submission.
GET/POST Incidents / Investigations	Initial report, triage, investigation and governed closure.
GET/POST Corrective Actions / Stop Work	Action create/Task conversion/verify; stop-work issue/release.
Workflow APIs	Submit/withdraw/respond/decide only through central Workflow where configured.

32. Events, Background Jobs, Audit and Observability

32.1 Event examples

```
hazard.created / risk_assessment.released  
permit.issued / permit.activated / permit.expired / permit.suspended  
inspection.failed / observation.reported  
incident.reported / incident.severity_changed / investigation.finalized  
corrective_action.overdue / corrective_action.verified  
stop_work.issued / stop_work.released  
worker.eligibility_changed  
emergency.activated / emergency.resolved  
environmental.limit_exceeded / environmental_event.reported
```

32.2 Background jobs

- Permit expiry and upcoming-expiry evaluation.
- Risk-review, action, induction/training and drill due/overdue scans.
- Worker/asset readiness recalculation after relevant source events.
- Dashboard aggregation/cache reconciliation and HSE metric snapshots.
- Search indexing, report/export generation and orphan-link integrity checks.
- Retries are idempotent, tenant-aware, correlation-aware and observable.

32.3 Audit / observability

- Actor/acting user, tenant/company/project/site, device/session, effective/server time, previous/new state, reason and correlation ID.
- Risk/RAMS/document/template/configuration versions used for controlled decisions.
- Permit activation/suspension/revalidation/closure and stop-work issue/release history.
- Incident evidence, investigation findings, witness access, reopen/addenda and sensitive reads where policy requires.
- Operational telemetry includes API latency, event backlog, failed integrations, cache freshness, permit/incident transition conflicts and offline sync failures.

33. Authorization, Privacy and Isolation

33.1 HSE role authority

- HSE Director / Manager / Officer / Investigator / Permit Issuer / Emergency Coordinator / Auditor capabilities remain distinct and scope-aware.
- HSE primary management authority does not replace HR, Assets, Inventory, QA/QC, Legal, Finance or Workflow authority.
- Stop-work release is a separately controlled action where policy requires.
- Every protected query, mutation, export, workflow action and sensitive read is backend authorized.

33.2 Protected HSE fields

Protected group	Examples
Medical	Medical detail, diagnosis, treatment detail beyond minimum operational projection.
Witness	Witness identity/statements and protected contact information.
Investigation	Root-cause analysis, confidential findings, investigator notes.
Legal/regulatory	Privileged legal strategy, protected regulatory response context.
External parties	Contractor/internal-only investigation detail and confidentiality markers.

33.3 Personal dashboard isolation

```
tenant_id + company_id + user_id + active_role_id + project/site_scope + contractor/date_filters + authorization_version
```

- Cache keys and precomputed dashboard summaries are isolated by effective authorization context.
- Revocation invalidates relevant client/cache projections.
- Cross-tenant, cross-company, cross-project and restricted-field leakage is a P0 security defect.
- Error messages and aggregate counts cannot reveal hidden sensitive records.

34. Responsive, Mobile, Offline and Accessibility

34.1 Responsive behavior

Desktop	Tablet	Mobile
Command dashboard, risk heat map, permit board, incident register, corrective-action board, split list/detail and export.	Adaptive list/detail, permit execution workspace, inspection checklist mode, incident evidence capture and signature panels.	Work Inbox/critical state first; report hazard/incident/near miss; issue stop-work; view permits; inspect; capture evidence; record induction/toolbox; verify eligibility; emergency access.

34.2 Offline-supported drafts

- Hazard, observation, inspection and dynamic risk-assessment draft.
- Initial incident/near-miss report and witness/evidence capture.
- Toolbox and induction attendance.
- Permit checklist preparation, not activation/closure.
- Corrective-action evidence and environmental readings.

34.3 Not finalized offline

- High-risk permit activation or closure.
- Stop-work release.
- Critical incident closure / final investigation.
- Risk-assessment release / HSE Plan issue.
- Emergency resolution or regulatory submission.

34.4 Accessibility and motion

- Critical status includes text/icon/accessible label; color is supplemental only.
- Keyboard, focus, screen-reader semantics, touch targets and responsive zoom meet platform accessibility baseline.
- Reduced-motion preference is respected; no decorative motion during emergency workflows.

35. Loading, Empty, Error and Conflict States

State	Required behavior
No authorized project/site	Show scoped empty state; never fall back to all company sites.
No critical items	Show clear safe/clear operational state without hiding data freshness.
Loading	Use skeletons while preserving layout; do not present stale values as current without freshness label.
Partial source failure	Healthy regions remain usable; failed region shows source unavailable/retry and clearly marks any stale cached value.
Access revoked	Remove protected content, clear stale client state and block deep-link access.
Permit activation gate failed	Show exact unmet gates; do not partially activate.
Version conflict	Explain source record/revision changed; reload/review before transition.
Offline conflict	Do not silently discard field evidence or overwrite newer server state; route to conflict review.
Workflow finalization failure	Decision may be approved but HSE transition remains pending/failed until source confirms.
Emergency concurrency	Enforce scenario rules for single/multiple active events; deterministic server resolution.

36. Performance, Testing and Release Quality

36.1 Performance targets

Area	Target / rule
Dashboard / page	First usable HSE dashboard/page under 2.5 seconds p75 under normal enterprise broadband conditions.
Normal lists	Indexed operational queries use server filters/cursor pagination and virtualized long tables.
Critical transitions	Permit activation, stop-work and incident actions require server validation and idempotency; no optimistic final state.
Evidence	Lazy thumbnails/signed URLs/resumable uploads; large originals do not block page load.
Scale	Designed for multi-company, multi-project, high-volume HSE activity and event history.

36.2 Critical test scenarios

- Risk revision while permit active preserves original reference and triggers revalidation rules.
- Stop-work suspends affected permit/work projections and governed release restores only validated scope.
- Unauthorized medical/witness/investigation access returns no sensitive payload.
- Task completes but HSE corrective action is ineffective: action reopens or additional action is created.
- Cross-tenant identifier enumeration returns no data.
- Offline hazard/incident evidence sync retries without duplicate critical records.
- Critical status remains understandable without color.
- Role-context shared-route tests keep HSE shell active.

37. Acceptance Criteria

1. HSE login opens the fixed HSE Dashboard and exact approved sidebar defined in this PRD.
2. Dashboard contains exactly the approved regions and eight primary cues.
3. Every cue, critical alert and readiness panel reconciles with permission-filtered authoritative source records.
4. Company/project/site/contractor/date filters narrow data without expanding authorization.
5. Immediate Attention remains highest priority and cannot hide active emergency/critical hold conditions through ordinary preference.
6. Today is site-time-zone aware and does not infer safe-to-start from schedule alone.
7. Risk/readiness calculations identify exact blockers, data freshness and source records.
8. Risk assessments preserve control hierarchy, immutable released revisions and explicit residual risk.
9. Expired/superseded/suspended RAMS blocks configured work/permit gates.
10. Permit activation executes all configured gates server-side and detects conflict/expiry/suspension correctly.
11. Stop-work immediately blocks affected work/permits and release uses governed separate authority where configured.
12. Inspection templates/results preserve historical revision/evidence and critical failures trigger configured consequences.
13. Observations support rapid mobile capture and immediate-danger escalation.
14. Incidents support initial report, triage, response, investigation, root cause, action, verification and governed closure.
15. Medical, witness and investigation fields remain independently permission restricted.
16. Corrective execution remains in Tasks while HSE owns verification/effectiveness/closure.
17. Worker readiness uses HR/asset/inventory source projections without copying masters or private medical detail.
18. Emergency readiness/state is source-controlled and critical active state remains visible.
19. Environmental/chemical records preserve HSE ownership while Inventory/Documents retain their own source truth.
20. Contractor HSE is scoped and evidence-based without hidden individual employee scoring.
21. Site HSE workspace consolidates projections without duplicating source records.
22. Workflow owns routing/decisions while HSE owns final source validation and state transition.
23. Shared Project/Task/Document/HR/Asset/QAQC/Calendar/Workflow routes preserve HSE shell and active role.
24. Platform-disabled features reflow cleanly with no dead links or blank regions.
25. Mobile/offline permitted/restricted actions behave exactly as defined.
26. Loading, partial failure, revocation, stale version and sync conflict states are explicit and safe.
27. Performance, large-data, accessibility, security and visual-regression targets pass.
28. No unapproved HSE dashboard widget or sidebar item is present.

38. Explicitly Not Allowed and Definition of Done

38.1 Explicitly not allowed

- Additional HSE dashboard KPI cards, charts, modules or shortcuts not defined in this PRD.
- User rearrangement of dashboard regions in v1.
- Automatic access to every company, project, site, contractor, incident or confidential investigation.
- Finance, Sales, HR productivity or unrelated company KPI widgets on the HSE Dashboard.
- Duplicating HR, Assets, Inventory, Tasks, Projects, QA/QC, Documents, Contracts, Finance or Workflow records in dashboard storage.
- Editing issued risk assessments, active/final permits, final investigations or closed HSE records in place.
- Treating PPE as equivalent to elimination, substitution or engineering controls.
- Activating permits client-side/offline or backdating activation.
- Releasing stop-work offline or through a generic administrator bypass.
- Treating Task completion as HSE corrective-action effectiveness/closure.
- Exposing detailed medical, witness or investigation content because a user can see an incident summary.
- Using safety metrics to automatically label individual workers as high/low performers.
- AI hazard/PPE detection, wearables, sensors, geofencing, telematics or regulatory integrations as required core behavior.
- Hard-coded role checks replacing platform capability/scope authorization.

38.2 Definition of Done

1. Exact HSE sidebar, header and fixed dashboard layout are implemented.
2. Eight primary cues and all approved dashboard regions reconcile with authoritative HSE/connected source records.
3. Hazard/risk/RAMS/permit/inspection/incident/action/stop-work/people/emergency/environment/contractor deep links open exact source records.
4. Site HSE workspace preserves project/site context and source ownership.
5. Permit activation, stop-work issue/release and incident closure pass end-to-end safety validation tests.
6. Workflow integration passes submit/decision/source-finalization, confidentiality and separation-of-duty tests.
7. Role-context persistence and permission filtering pass direct-link/search/notification/shared-route tests.
8. Responsive/mobile/offline/accessibility/loading/error/conflict states pass visual and functional QA.
9. Performance and cache-isolation targets pass with representative enterprise data.
10. Audit, immutable-history and negative isolation tests pass with zero critical defects.
11. Dashboard scope lock is respected with zero unapproved elements.

Final implementation rule

If an element is not defined in PRD_HSE_Dashboard, it does not appear on the HSE Dashboard or HSE role sidebar. New dashboard content requires an approved PRD revision.

Appendix A. Source Alignment Matrix

PRD decision	Primary source alignment
HSE ownership, information architecture and operational domain	PRD_HSE_Module v1.0 + PRD_HSE v1.0.
Dashboard immediate attention/today/leading/lagging/readiness behavior	PRD_HSE_Module HSE Dashboard + PRD_HSE dashboard metrics.
Exact role-center structure and Work Inbox	PRD_2 Role Centers + Platform UI/UX Architecture.
Hazards/risk/control hierarchy	HSE source PRDs risk architecture and control-hierarchy requirements.
RAMS and method readiness	PRD_HSE RAMS structure, lifecycle and work-gate behavior.
Permit lifecycle and activation gates	HSE source PRDs Permit-to-Work / LOTO architecture.
Inspections/observations/incidents/investigations/actions/stop work	HSE source PRDs field-control and incident-response architecture.
People readiness / HR boundary / PPE	HSE source PRDs + Complete Role & Permission Matrix.
Emergency/environment/chemicals/contractor HSE	HSE source PRDs approved module scope and phases.
Approval routing / emergency override / source finalization	Approvals & Workflow Engine v1.0.
Role authority, sensitive medical/witness/investigation data	Complete Role & Permission Matrix v1.0.
Shared role shell behavior	Role Context and Shared Module Navigation.
Responsive/mobile/accessibility/progressive disclosure	Platform UI/UX Architecture + HSE mobile/offline rules.

Appendix B. Route and Context Map

```

/hse
-> /hse/sites / sites/{id}
-> /hse/plans
-> /hse/hazards / risk-assessments / RAMS links
-> /hse/permits / permits/{id}
-> /hse/inspections / observations
-> /hse/incidents / investigations / actions
-> /hse/stop-work
-> /hse/inductions / toolbox-talks / competence / ppe
-> /hse/emergency / environment / chemicals / contractors / occupational-exposure
-> /hse/reports / settings

Shared source routes
-> Project / Work Progress / HR / Asset / Inventory / Task / QA/QC / Document / Contract / Procurement /
Calendar / Government & Legal / Finance / Workflow
-> active HSE role shell remains resolved from session context

```

Final architectural position

HSE is an active operational safety gate, not a retrospective reporting screen. HSE owns whether safety controls, permits, holds, incident response and environmental controls are valid. Work Progress owns physical execution, QA/QC owns quality acceptance, HR owns people/private health data, Assets owns equipment lifecycle, Documents owns files and Workflow owns decision routing. The HSE Dashboard presents these authoritative states without duplicating ownership or allowing a favorable score to conceal a mandatory safety blocker.